To: Refugee Researchers at the Immigration Policy Lab

From: Evgeniia Buzulukova, Yufei Xiao

Subject: Improving dimensionality and heterogeneity in survey data

Our replication and extension code are in this repository:
https://github.com/yufeixiao/Replication.

Abstract:

Refugee policymakers are interested in the causal question: what factors shape attitudes toward refugees in host countries? In this paper, we examine Hangartner et.al. (2019)'s instrumental variable case study, replicating their causal effect estimates of exposure to refugees on local hostility. Furthermore, we demonstrate that their analysis is marred by missing data; by performing proper imputation and dimensionality reduction procedures and contextualizing their model choice, we found differing conclusions that may even reverse the direction of effect of exposure on attitude.

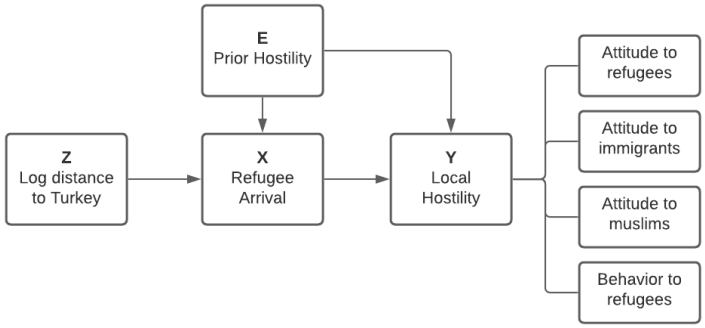
Introduction:

We are replicating and extending Hangartner et.al. (2019), an instrumental variable case study. They collected survey data (N=2046) from residents of Greek Islands during the 2016 Syrian refugee crisis.

This study is important because the direction and magnitude of causal effect can lead to inference about the mechanism behind how interaction with refugees affects resident hostility and shifts the political atmosphere.

The authors perform a two-stage least-squares regression to estimate the treatment effect of refugee arrival on indicators of resident hostility, with distance to Turkey as an instrumental variable. Using the instrument the authors can ignore the endogenous interaction, which is the refugees choosing to go to places that are already less hostile.

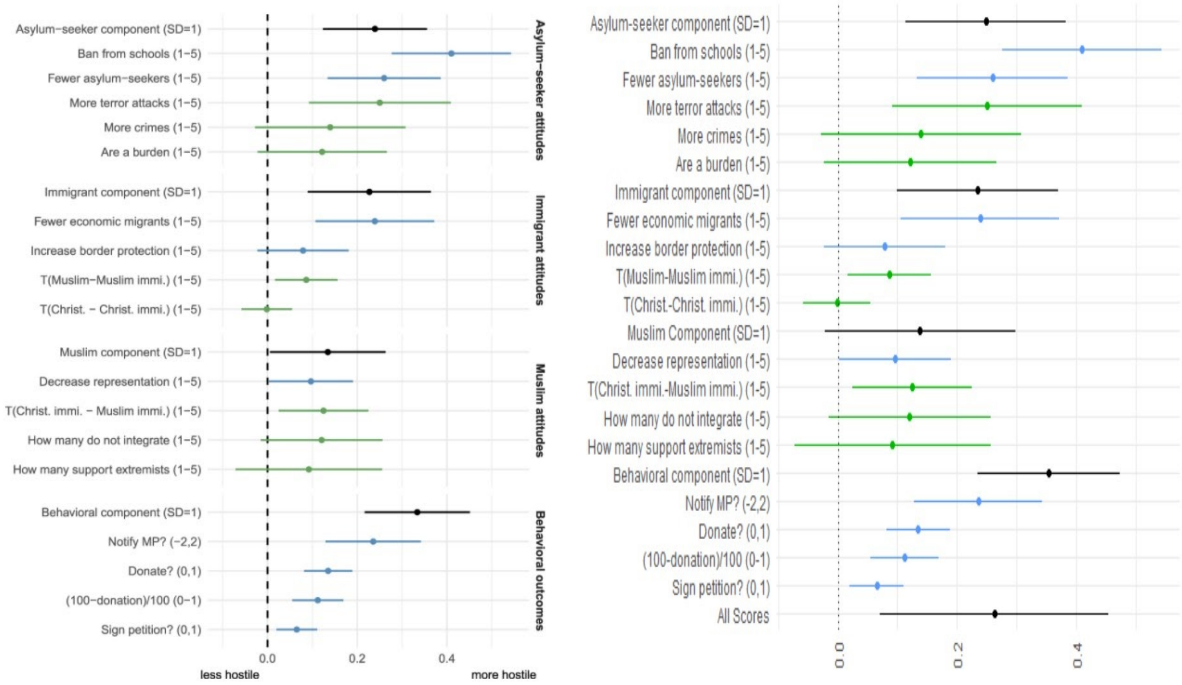
$$T = a + bZ + u$$
$$Y = c + dT + v$$



[Figure: Causal graph]

Replication: #studyreplication

We were able to replicate Figure 4, which shows causal effect estimates for four categories of responses and their first principal component, to full precision.



[Figure: original (left) vs replicated (right) instrumental variable regression estimates]

After replication, we found important information not presented in the results: 15006 out of 61560 (24.4%) principal component scores are missing after the PCA procedure, even though a majority of the points have sufficient data for computing a principal component.

Why is their method for calculating PCA missing some calculations even though they already have the sub-scores? We could not find any mention of the large missing data, as authors claim to have imputed all missing outcome values and found all PCA scores. Dropping such a significant portion of data is problematic, because it likely makes the analysis less generalizable to the whole population.[1]

Extension Methods:

Our minimal improvement is changing the way the author dealt with missing response data. Since our goal is to approximate the unobservable constructs like attitude toward immigrants, we reduce bias from the highly dimensional and heterogeneous survey data using statistical methods like imputation and dimensionality reduction.

We fill in the missing data by Multiple Imputation using Chained Equations (MICE). By assuming that missingness is random, we can salvage more data points by predicting missing values using covariates.

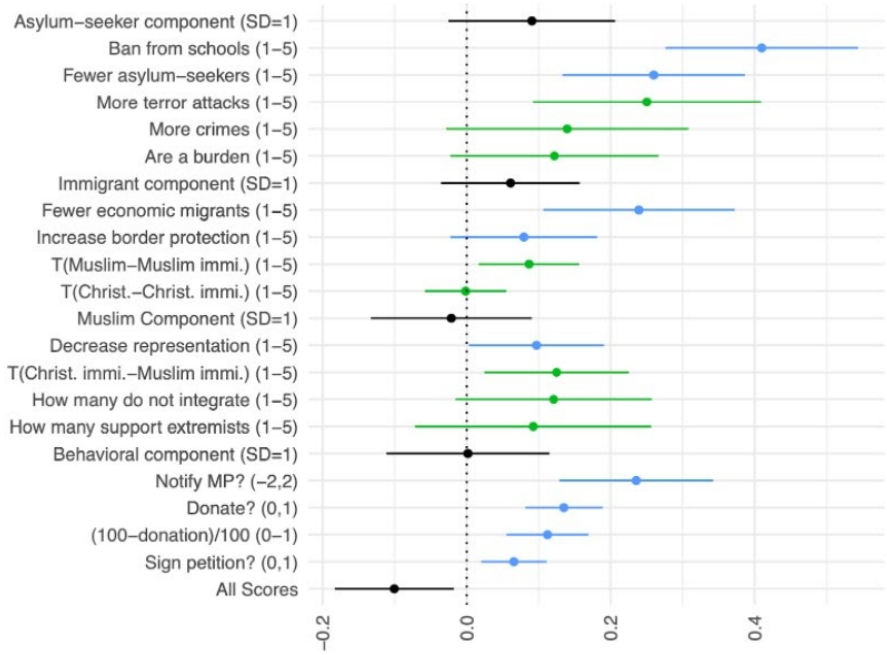
We compared different dimensionality reduction methods for generating summary scores to see how the original study's choice, PCA, affects the shape of results. In addition to Principal Component Analysis (PCA), we tried two popular nonlinear methods: t-stochastic neighbor embedding (tSNE) and uniform manifold approximation and projection (UMAP).

When choosing between methods, we need to examine fit to the particular data structure and computational budget. PCA is fast and deterministic but assumes a strict normal distribution, whereas UMAP can preserve global nonlinear patterns but is stochastic and can become computationally expensive. Given our high dimensional survey data, we might expect complex interactions between variables, which might benefit from the flexible nature of multimodal models.

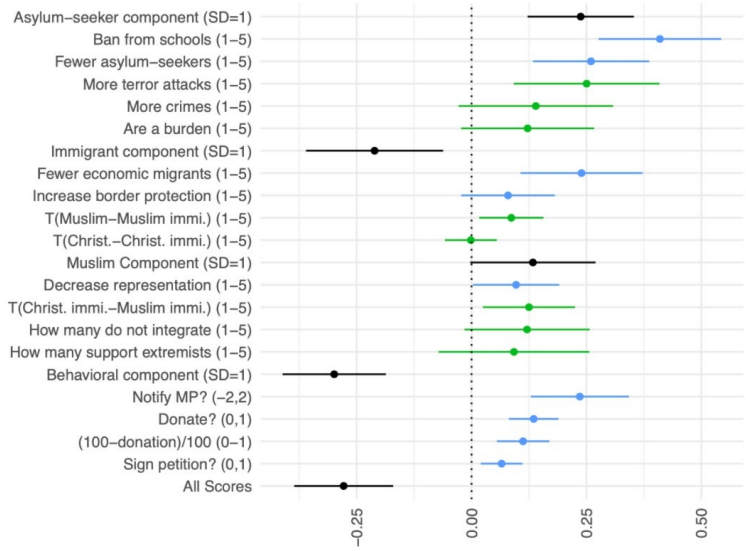
Extension Results: #biasmitigation

From the causal estimate results we can see that our methodological change caused substantive conclusion change.

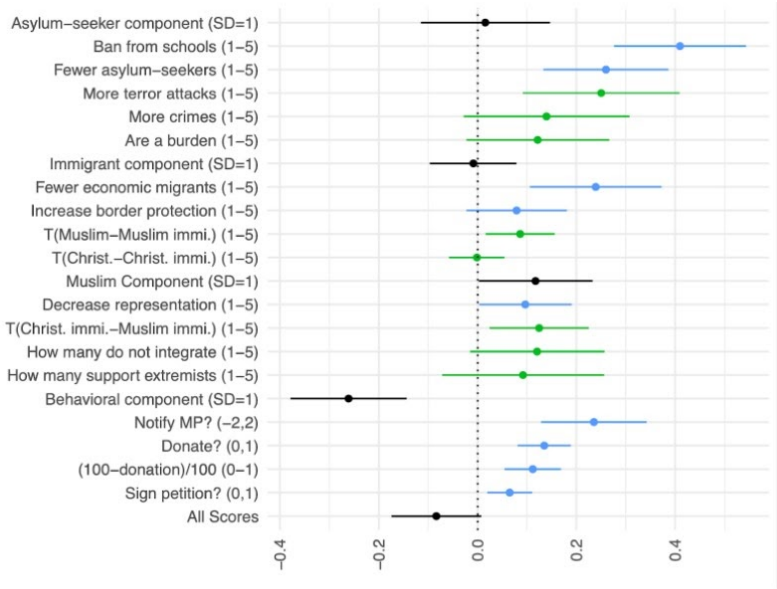
[1] #studyreplication: we explicitly state precision of replication, and later use the same graphic scheme to compare old conclusions with new ones from extensions, following King, Tomze, Wittenberg (2000)'s advice to incrementally improve results presentation.



PCA estimates with imputation



tSNE estimates with imputation



UMAP estimates with imputation

Trying different methods and checking if their results agree lets us reduce selection bias by depending less on model assumptions. A consistent difference from the original conclusion might indicate either valid new conclusion or sensitive data. This difference is meaningful even if we are not right, because it reveals properties about the data.

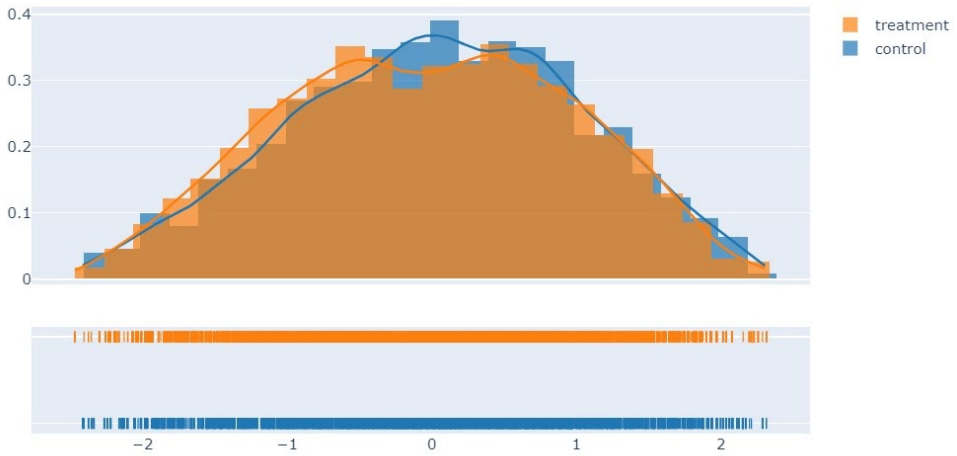
We test the correlation between these dimensionality reduction methods. The Pearson R between PCA and tSNE is 0.34 (only calculated for the first component), suggesting that the components generated by each method are positively correlated. However, the correlation between UMAP and TSNE or UMAP and PCA are not presented as Pearson R is about 0. Lack of agreement implies that we cannot assume functional reasoning behind the boundaries they create.

Firstly, we used 3 different metrics but got similar results that consistently deviate from the original study's estimates. As we can see from the figures replicated using our estimates, most of the scores started to have a negative correlation with the treatment. It means that the imputations and correct dimensionality reduction (without having the unexpectedly missing values in the score metrics) led to a very different conclusion from the original study. If we were right, refugees arriving on an island make the locals more sympathetic, especially in their voting behaviors regarding voting and donation.[2]

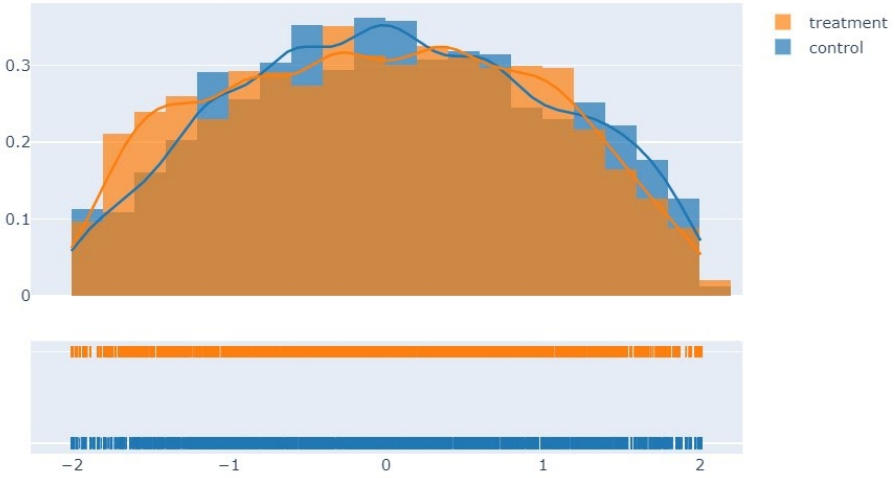
Secondly, the overall pattern is sometimes opposite in causal effect from the subcategories. Why is it so and what does it mean in this case? It could be Simpson's Paradox. Slicing into the subcategories may have introduced noise from the survey data's self-reported scales and correlated questions, which influences both the treatment and the outcomes.

[2] #biasmitigation: contextualizing a model choice reduces selection bias

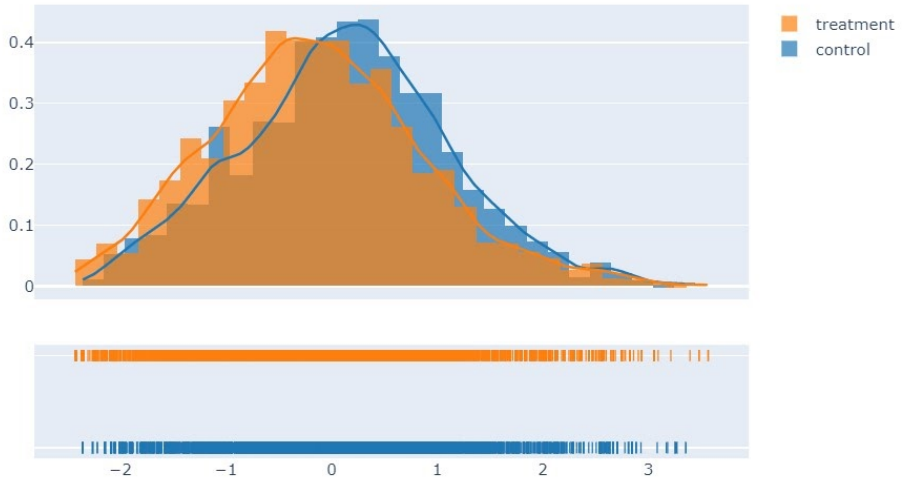
We also show the distributions of all scores combined for treatment and control groups. Since we only have one component (as specified by the authors, who argue we only need the component with largest explained variance), this is our visualization of clustering by each metric.



PCA distributions



tSNE distributions



UMAP distributions

We cannot choose a dimensionality reduction method by maximizing the separation between groups, because by principle the outcome should not depend on the assignment. However, the differing shapes of these distributions tell us about the assumptions beneath each metric. For example, UMAP's higher kurtosis and positive skew might suggest a more prominent grouping.

Recommended Analysis:

The limitations of our analysis include having no measure of accuracy, since we are processing data for a causal rather than predictive problem; assuming instrument validity without checking covariate balance.
To quantify the randomness of missing data, we might analyze the covariate balances more to profile compilers and non-compliers to estimate how much our local average treatment effect can generalize to the population.
Following Hangartner et.al.'s groundbreaking causal study, refugee researchers can adopt a similar set of study design and statistical methods. They can refer to our modifications on handling missing data and choosing score models.

Conclusion: #emergentproperties

We have shown that Hangartner et.al. (2019)'s causal effect estimates for component scores of resident hostility due to refugee arrival suffer from mishandled data filling procedures. We produce estimates for several correction methods to contextualize the original study's results and observe a general reduction in the effect of arrival than the original claim, with one component (behavior) consistently reversing the direction of effect. That means, compared to what Hangartner said, having refugees arrive on an island doesn't cause as dramatic of increased hostility toward refugees, immigrants, Muslims, and it may even decrease hostile behaviors.

Whether or not our conclusion is right, what theoretical implications are there for the Immigration Policy Lab researchers? The dynamics of resident sentiment toward refugees -- from hostility to empathy-- emerges from complex interaction between and within each group and with the intermediate policy-enacting agencies. Working hypotheses about this emergent phenomena mainly spring from migration theory and involve the paradoxical loops between contact between groups reducing misconceptions and heating clashes. This study attempts to capture such a dynamic through dimensionally reduced attitude scores, which is a continued challenge for future studies.[3]

Do findings of what drives public opinion in Western countries apply to countries like Jordan? We would expect interruption to daily life, perceived religious difference and economic burden to play out similarly in a different country. There could be assumptions underlying studies in Western countries that would get exposed to a study in Jordan.

References

1. Gary King. 2006. "Publication, Publication." PS: Political Science and Politics, 39, Pp. 119–125. Continuing updates to this paper Copy at https://j.mp/2owxpXM

2. Hangartner, D., Dinas, E., Marbach, M., Matakos, K., & Xefteris, D. (2019). Does Exposure to the Refugee Crisis Make Natives More Hostile? American Political Science Review, 113(2), 442-455. doi:10.1017/S0003055418000813

3. Marbach, M., & Hangartner, D. (2020). Profiling Compliers and Noncompliers for Instrumental-Variable Analysis. Political Analysis, 28(3), 435–444. https://doi.org/10.1017/pan.2019.48

[3] #emergentproperties: explain the challenge with modeling public opinions

Clarify with precision the extent to which you were able to replicate the author's results.

We were able to replicate the entire results using their data and code. Particularly we replicated figure 4 in the original study, which shows the confidence intervals of causal effect estimates of refugee arrival on local attitude, policy preference, behavior.

If you decide that the conclusions of the original article are incorrect, then show why you think that but also what led the authors of the original article to think otherwise.

After replicating the article, follow the logic of King, Tomz, and Wittenberg ~2000! and try to improve the presentation of the original results. See whether you can find useful, additional, or even contradictory information not discussed in the article without changing any assumptions in the original paper.

That is, make one improvement, or the smallest number of improvements possible to produce new results, and show the results so that we can attribute specific changes in substantive conclusions to particular methodological changes. ~Improvements can include **changing the way the author dealt with missing data**, selection bias, omitted variable bias, the model specification, differential item functioning, the functional form, etc., adding control variables or better measures, extending the time series and conducting out-of-sample tests, applying a better statistical model, etc.!

- Our minimal improvement is changing the way the author dealt with missing data.
- From the results we can see that our methodological change caused substantive conclusion change.

0. Provide evidence that your model fits the data in and out of sample ~or perhaps that it fits better than the model in the original article!

- Our confidence intervals are narrower.
- We did not have a set-aside dataset to test prediction accuracy.

To: Immigration Policy Lab

From: Evgeniia Barabush, Yufei Xiao

Subject: Modified findings about the effect of refugee exposure on local hostility

Executive Summary

Arsenal attacks on refugee shelters, anti-refugee protests on the streets... What drives public opinion against refugees? Policymakers supporting refugees look for explanations and effective interventions.

To understand how refugee handling affects resident attitude, Immigration Policy Lab researchers can learn from the methods and conclusions from Hangartner et.al. (2018), the first causal study about refugee exposure on resident attitude, from the Syrian refugee crisis in 2016.

To verify conclusions about the effect of refugees arriving on resident hostility, we replicate and extend Hangartner et.al. (2019)'s landmark paper on 2016 refugee crisis ("Does Exposure to the Refugee Crisis Make Natives More Hostile?").

Our minimal improvement is changing the way the author dealt with missing data.

We identify statistical analysis problems in their missing data and dimensional reduction procedures which reduce their causal inference reliability, then correct them and compare the differing conclusions and find that effects generally reduced. Our replication and extension contextualizes the method choices made by the authors and gives more reliable estimates.

Background

This is a causal inference problem, and we don't know the real causal effect. We cannot decide which statistical method is better. However, the difference in the causal estimate effects show us the limitation of one method and the underlying structure of the dataset.

What if refugees choose islands because they are less hostile? This is the endogeneity problem we want to fix. We are concerned that there are still invisible things skewing our conclusion about whether accepting refugees is making locals reject refugees more.

The authors use a two-stage least square regression to estimate the treatment effect of refugee arrival on indicators of resident hostility with distance to Turkey as an instrumental variable.

For missing values, authors claim to mark missing values for covariates with a binary indicator, and use Multiple Imputation by Chained Equations (MICE) to impute missing values for outcome variables. MICE assumes that data are missing at random, so their values can get

predicted based on observed values. Then the authors create four broader components (asylum seeker attitudes, immigrant attitudes, Muslim attitudes, behavioral outcomes) to reduce false positive rate that comes from too many outcomes. They claim to use Principal Component Analysis to find the first principal component for the entire dataset.

Replication #studyreplication

In terms of objective, study wants to investigate the effect of refugees arriving on the attitude, policy preference, behavior of local residents of Greek islands during the 2016 refugee crisis. This is important because the direction and magnitude of causal effect can lead to inference about the mechanism behind how interaction with refugees affects resident hostility and shifts the political atmosphere.

Maybe the instrument is not so good. For example, two islands at the same distance from Turkey can have very different attitudes because of their cultures. Matching can fix covariate balance better than a single instrument.

There is a little missing data in covariates but there are a lot of missing data in PCA scores.

Firstly, 2,776 out of 221,616 (1.25%) data points are missing outcome values even after the imputation procedure was passed.

Secondly, 15,006 out of 61,560 (24.4%) principal component scores are missed even after the PCA procedure was passed. Upon further checking, we conclude that a lot of points do have outcome values that contribute to the missing components, hence there is no reason to leave them out of the PCA calculation.

These two facts are problematic because (1) leaving out 24.3% of the dataset increases variance and reduces generalizability, (2) we cannot assume values are missing at random, and the more we leave out, the more risk of biasing our results. Hence, the reliability of the causal effect direction and magnitudes come into question.

The impact is that it makes really big confidence intervals in the original causal estimates.

Why is their method for calculating PCA missing some calculations even though they already have the sub-scores?

Extension #biasmitigation

PCA

tSNE

UMAP

PCA

tSNE

UMAP

- data imputation
- dimensionality reduction

We extend the study by addressing missing data. Our motivation is to correct the missing values and dimensional reduction and see how the results change. Building on the same approach of instrumental variable binary treatment multiple regression, first we modify missing data methods: we impute missing outcomes. Then, we compare imputation with dropping. Second, we modify dimensional reduction methods: we include all data in PCA. Then, we compare PCA with tSNE and UMAP.

We try different dimensionality reduction methods to see if the original study's choice, PCA, is limited.

Dimensionality reduction on the survey responses can decrease the chance of false positive tests. We cannot know how faithfully the composite scores render real latent constructs of attitude and behavior, because they are by definition unknowable (in unsupervised learning), but we can try to use methods whose assumptions and results fit with the structure of our data.

When choosing between methods, we need to look at pros and cons. UMAP can cluster geometrically, is better with nonlinearities, but it's stochastic. tSNE doesn't preserve the local structure and works. PCA is usually faster but is stricter.

Discussion #emergentproperties

- difference in causal estimates
- sensitivity

We test the correlation between these dimensional reduction methods. The Pearson R between PCA and tSNE is 0.34 (only calculated for the first component), suggesting that the components generated by each method are positively correlated. However, the correlation between UMAP and TSNE or UMAP and PCA are not presented as Pearson R is about 0. Lack of agreement implies that we cannot assume functional reasoning behind the boundaries they create.

Firstly, we used 3 different metrics (DR) but got similar results that consistently deviate from the original study's estimates. As we can see from the figures replicated using our estimates, most of the scores started to have a negative correlation with the treatment. It means that the imputations and correct dimensionality reduction (without having the unexpectedly missing values in the score metrics) led to a very different conclusion from the original study.

Secondly, the overall pattern is opposite in a causal effect from the subcategories. Why is it so and what does it mean in this case? It could be Simpson's Paradox: the overall behavior score

Another explanation is that our imputations might have wrong assumptions: if which data is missed is not random, the original data structure is not preserved after filling and including the missing values. Maybe it's because we are predicting from covariates but in fact missing is not random. We don't know why the authors dropped the missing values in their PCA calculation.

Supposing we are right, what theoretical implications are there for the Immigration Policy Lab researchers? The dynamics of resident sentiment about refugees and immigrants is an emergent property because it is a crowd mood generated from residents influencing each other on individual and group levels. This study attempts to capture such a dynamic through dimensionally reduced self-reports, which is an honorable attempt to measure such a complex latent construct.

However we cannot be sure who is right, but we can offer possible reasons and how to correct for them.

The response data is so sensitive to change in methods and mistakes. The authors did not address why these particular PCA scores are missing. Perhaps the data is not missing at random.

Another possible flaw of this study is, the instrument is not strong enough. For example two islands at the same distance from Turkey can have very different attitudes. we can check instrument validity. We want to check if there is an imbalance of an unobserved characteristic between the treatment and control groups which may bias the results, and potentially use matching to balance covariates.

In the future, we can improve the study design and the method design by collecting more data that has less missing values or has more empirical evidence to support assumptions behind each missing data fix and score-summary fix. For example, the covariate balance of respondents and non-respondents can support the assumption about random missing data.

Conclusion

We have shown that Hangartner et.al. (2019)'s causal effect estimates for component scores of resident hostility due to refugee arrival suffer from mishandled data filling procedures. We produce estimates for several correction methods to contextualize the original study's results and observe a general reduction in the effect of arrival than the original claim, with one component (behavior) consistently reversing the direction of effect. That means, compared to what Hangartner said, having refugees arrive on an island doesn't cause as dramatic of increased hostility toward refugees, immigrants, Muslims, and it may even decrease hostile behaviors.

Do findings of what drives public opinion in Western countries apply to countries like Jordan? We would expect interruption to daily life, perceived religious difference and economic burden to play out similarly in a different country. There could be assumptions underlying studies in

Western countries that would get exposed a study in Jordan. Researchers can take the study design and statistical methods we modified to improve on causal inference on survey data there.

Appendix

Our replication and extension code are in this repository: https://github.com/yufeixiao/Replication.

References

Hangartner, D., Dinas, E., Marbach, M., Matakos, K., & Xefteris, D. (2019). Does Exposure to the Refugee Crisis Make Natives More Hostile? American Political Science Review, 113(2), 442-455. doi:10.1017/S0003055418000813

#emergentproperties: attitude score

#decisionbrief: include the question in the executive summary. Explain simply

#decisionquestion: causal inference problem

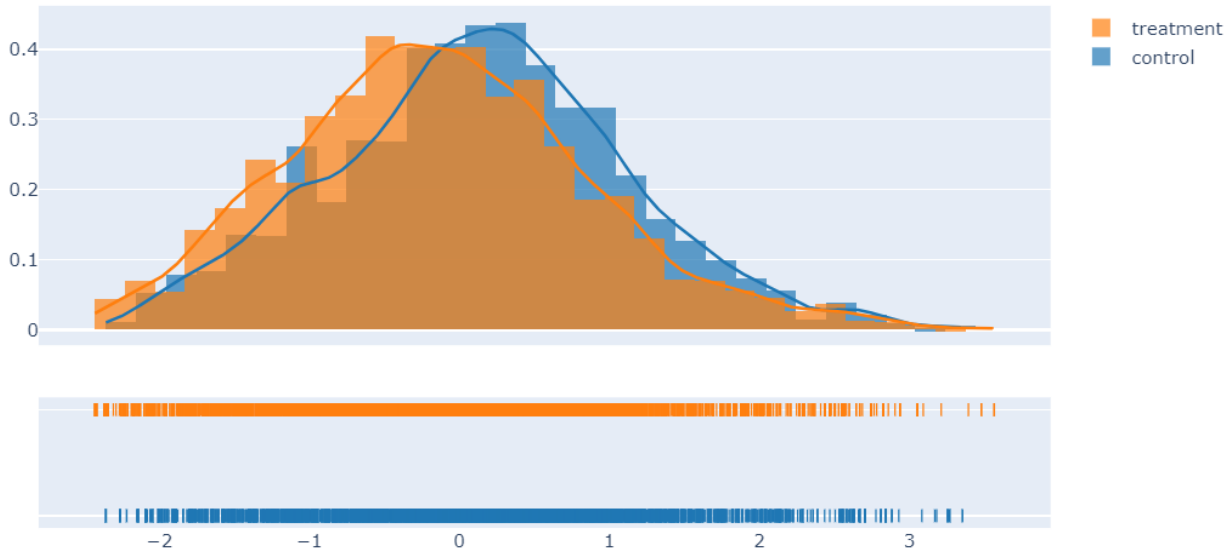
#decisiondesign: pros and cons of reduction for

- Clustering algorithms make assumptions about the structure of the data. What is the underlying distribution and what do we need the reduced dimensions for? PCA assumes normal distribution. Survey data probably has sampling bias and self-report bias.

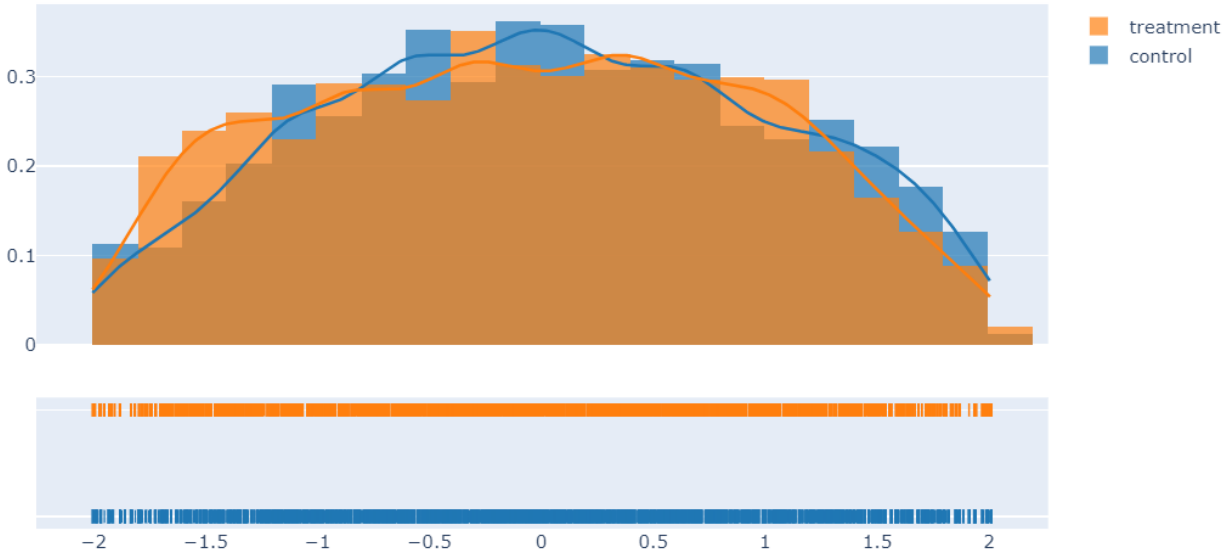
#biasmitigation: selection bias in missing survey data

#studyreplication: improve on methods using the same data

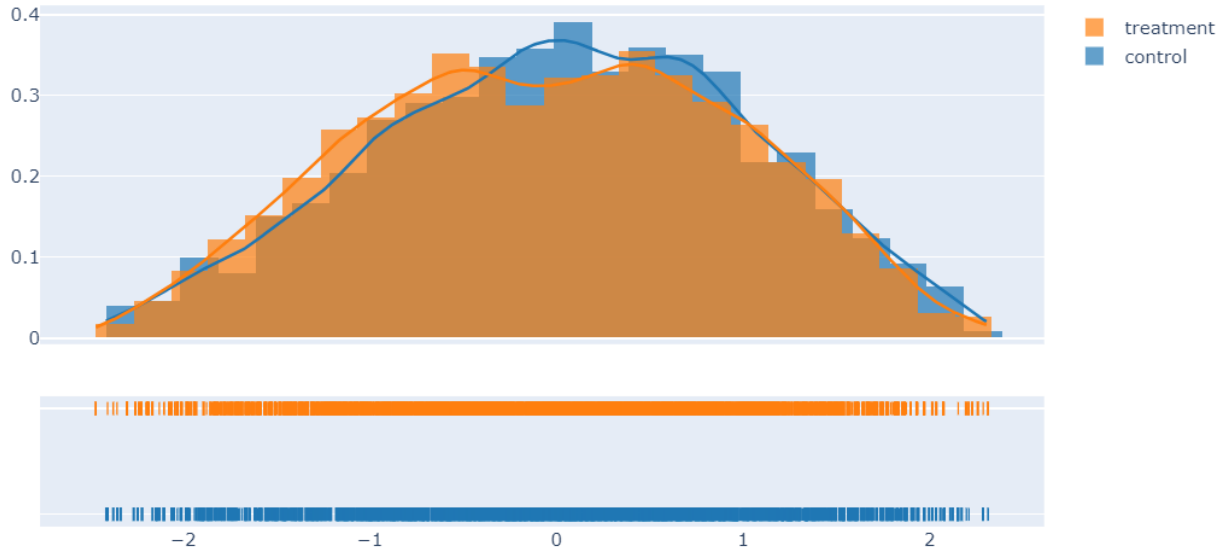
We tackle messy survey data to figure out why people repel refugees.



PCA all



tSNE all



UMAP all

#interviewsurvey: data have lot of outcomes which is vulnerable to false positive test
#casestudy: using the case of Greek islands to provide guidance to countries experiencing refugee crisis by seeing what policies and factors can help refugee and local get along.
#analogies: transferring the Greek island example
#regression: we are contextualizing our causal effect model by passing in data preprocessed in different methods to hopefully unbias from wrong assumptions
#biasmitigation: selection bias in nonresponse
#emergentproperties: hostility emerges from situated interactions between locals and refugees

Decision Recommendation

- Understanding resident reaction to refugees can inform a government to (1) make policy changes to drive down hostility by handling arrivals more smoothly and (2) increase interaction rather than exposure to help with refugee integration.
- A humanitarian organization can use our study and method recommendations to learn more about how interaction between refugees and locals affect the community's attitude about the refugee crisis.

Instrumental Variable Analysis

- Having a continuous instrument variable
- Concerns with the covariate balance even after instrument

There is a little missing data in covariates but there are a lot of missing data in PCA scores. That makes really big confidence intervals.

- What if refugees choose islands because they are less hostile? This is the endogeneity problem we want to fix. We are concerned that there are still invisible things skewing our conclusion about whether accepting refugees is making locals reject refugees more.
- Why are the results different?
    - Maybe the results are sensitive to confounding. That would make the dimensional reduction exaggerate differences
    - They are so sensitive to change in methods and mistakes. Perhaps the data is not missing at random. Why are these particular PCA scores missing?
    - Maybe this means the method design has flaws.
        - Why are these particular answers missing?
        - Why is their method for calculating PCA missing some calculations even though they already have the subscores?
- Maybe the instrument is not so good. For example two islands at the same distance from Turkey can have very different attitudes. Maybe we should match.
    - This is the study design.

2-Stage Least-Square Regression

Data Imputation

Dimensionality Reduction

- We try different methods to see if the original study is limited.
- We cannot decide which one is better. The difference in the causal estimate effects show us the limitation of one method.
- We are not choosing based on which one gives
- This is a causal inference problem, and we don't know the real causal effect.
- When choosing between methods, we need to look at pros and cons. UMAP can cluster geometrically, is better with nonlinearities, but it's stochastic. tSNE doesn't preserve the local structure and works. PCA is usually faster but is stricter.
- Resampling probably won't improve results because we would use it on stochastic methods which would turn out more differing outcomes.

Principal Component Analysis

- It's ok that we might not be right. Maybe we are wrong but it is possible that we are right because of these reasons.
- Why is the overall pattern opposite in causal effect from the subcategories? It could be Simpson's Paradox.
- The sub-outcomes retain the CI after imputation. That is good.
- Maybe the very different looking CIs are because our imputations are wrong - the original data structure is not preserved after filling and including the missing values. Maybe it's because we are predicting from covariates but in fact missing is not random. We don't know why the authors dropped the missing values in their PCA calculation.
- We used 3 different metrics (DR) but got the same result that consistently deviates from the original.

- Maybe there are issues with the original work. We did DR correctly.

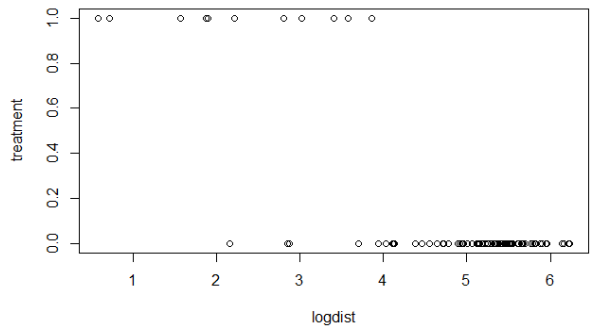
T-distributed Stochastic Neighbor Embedding

Uniform Manifold Approximation and Projection

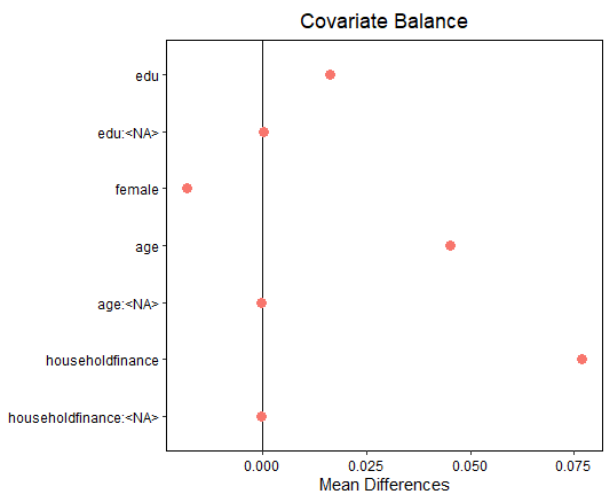
Appendix

Action items:

1. Plot covariate balance, show instrument is not balanced, try DR



a.




```
> MatchBalance(treatment~logdist,data=survey_data)

***** (V1) logdist *****
before matching:
mean treatment........ 1.9591
mean control.......... 5.2357
std mean diff......... -451.86

mean raw eQQ diff..... 3.2762
med  raw eQQ diff..... 3.3497
max  raw eQQ diff..... 3.797

mean eCDF diff........ 0.53005
med  eCDF diff........ 0.57115
max  eCDF diff........ 0.92918
```

b.

	GD	S/A	ND	PASOK	Female	Age	Education
treatment	0.002	-0.053	0.050	0.001	-0.020	0.007	-0.016
	(0.008)	(0.023)	(0.021)	(0.012)	(0.028)	(0.028)	(0.022)
constant	0.025	0.371	0.191	0.087	0.561	0.477	0.450
	(0.005)	(0.032)	(0.020)	(0.011)	(0.016)	(0.019)	(0.018)
N	2046	2046	2046	2046	2046	2046	2046

Table 1: Placebo Outcomes (Table 1 in original paper)

   c.
2. Plot outcome distributions, can't assume Gaussian, try nonlinear DR
   a. Behavioral is obviously multimodal
3. Print confidence intervals, compare effect sizes
   a. It can be a future extension to compare effect sizes, significance between the estimates.
4. Interpret refugee policy
   a.

How to know if a DR is accurate or now?

We should choose DR based on how well it separates between the treatment and control groups. That is biasing the outcome to become more extreme. We should choose DR based on the nature of the dataset, based on how balanced the covariates are, what assumptions we can make about the distributions.

Should we examine covariate balance to justify DR choice

Are the tr and ct separated in clusters in the graph?

Yufei write explanation

Can we evaluate covariate balance and classification accuracy in stage 1 of regression? (instrument on treatment)

Can we bootstrap the 2SLS

Can we describe the causal effects in terms of significance and confidence intervals

What does DR tell us about data pattern?

Low correlation between methods implies no reliable patterns

Do we need to resample for TSNE and UMAP

PCA assumes Gaussian distribution. tSNE assumes multimodal distribution. If we resample we should check if the new dataset satisfies those assumptions.

Unlike the deterministic PCA, tSNE is stochastic, there are many minima, results are less reproducible -- unless we use random seed. If rerunning tSNE gets us a difference

small enough then the divergence is meaningful. Because tSNE describe local trends it is less interpretable.

Decision memo for a country experiencing refugee crisis

- Replicate original figure 4
    - Yufei - IV & figure 4
        - Two-stage least square regression
        - $T = a + bZ + u$
        - $Y = c + dT + v$
    - Is it in R already?
    - Explain how they did IV regression
    - They used dimensional reduction (PCA) and multivariate regression (IV).
- Brainstorm extension
    - Evgeniia - any testable dimensional reduction methods? Do dimensional reduction
    - Survey data is likely nonlinear.
    - PCA to generate features only uses linear interactions to combine parameters. Alternatives: tSNE, kernel PCA, isomap, MDS. It can cluster better because it weights all elements, but it might not be able to reduce random measurements.
    - We cannot evaluate the accuracy of unsupervised learning, but we might be able to use "explained variance." tSNE doesn't have explained variance. Can we plot the clusters for treatment vs control units against dimensions?

- Other extensions in the future
    - Matching
    - Contextualize LATE in complier vs noncomplier populations using Hangartner's new software.

DECISION MEMO
TO: Colombian government
FROM: Evgeniia Barabush, Yufei Xiao
SUBJECT: How Refugee Arrival Impacts Local Politics
EXECUTIVE SUMMARY:
Colombia accepts Venezuelan refugees fleeing from pandemic, economic instability, climate change, conflict. It is important to understand what is driving anti-refugee sentiments, because the Colombian government wants to integrate refugees into their society. If indeed even exposure increases hostility, then the government needs to improve arrival handling to avoid these prejudiced policies. The government should also support big data studies to help the acceptance and integration of refugees. Huma

To understand how refugee handling affects resident attitude, investigators from the Colombian government can learn from the methods and conclusions from Hangartner et.al. (2018), the first causal study about refugee exposure on resident attitude, from the Syrian refugee crisis in 2016.

To verify conclusions about the effect of refugees arriving on resident hostility, we replicate and extend Hangartner et.al. (2019)'s landmark paper on 2016 refugee crisis ("Does Exposure to the Refugee Crisis Make Natives More Hostile?"). We identify statistical analysis problems in their missing data and dimensional reduction procedures which reduce their causal inference reliability, then correct them and compare the differing conclusions and find that effects generally reduced. Our replication and extension contextualizes the methods choices made by the authors and gives more reliable estimates.

REPLICATION:
In terms of objective, study wants to investigate the effect of refugees arriving on the attitude, policy preference, behavior of local residents of Greek islands during the 2016 refugee crisis. This is important because the direction and magnitude of causal effect can lead to inference about the mechanism behind how interaction with refugees affects resident hostility and shifts the political atmosphere.
In terms of statistical methods, the study draws causal inference from non-interventional data using instrumental variable analysis.
The authors use a two-stage least square regression to estimate the treatment effect of refugee arrival on indicators of resident hostility with distance to Turkey as an instrument.
Loading…
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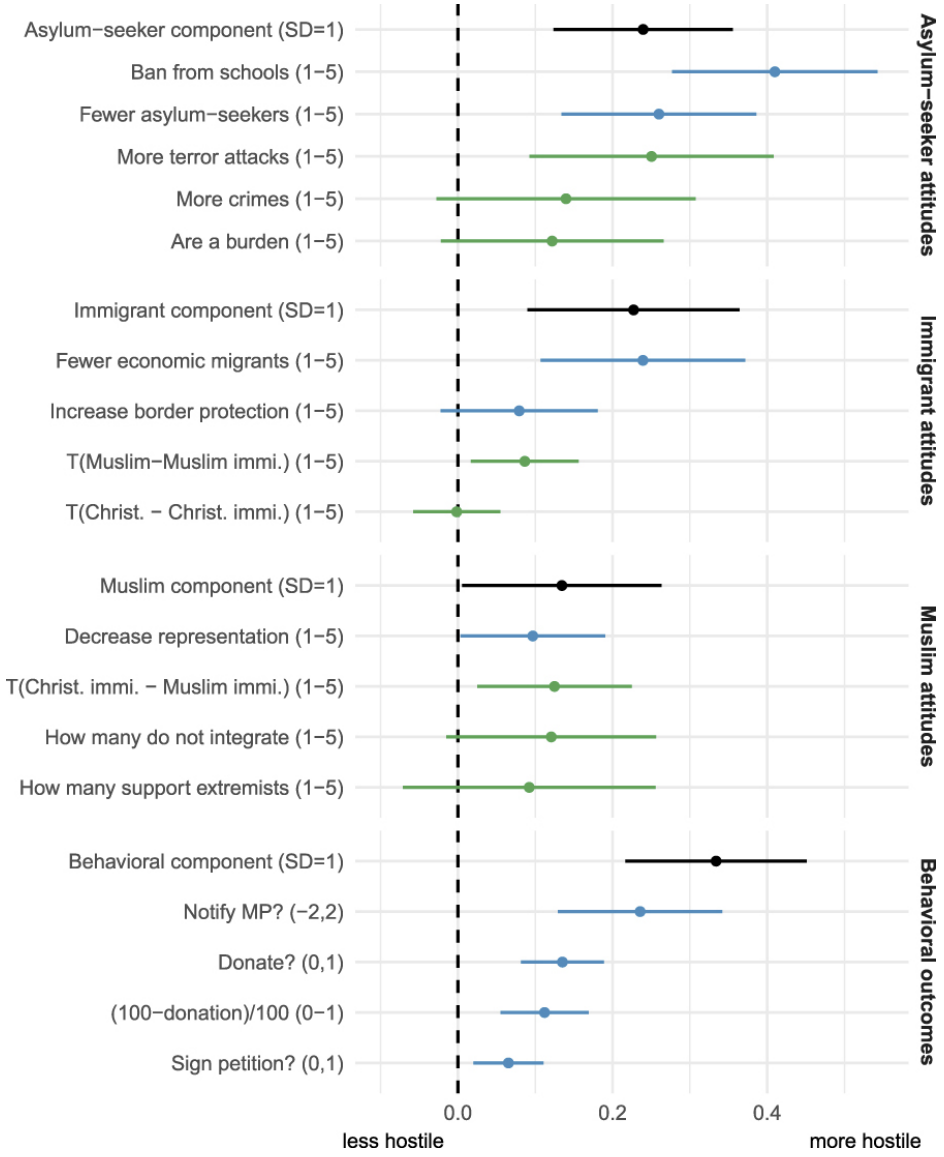
Where: Each unit is a resident on a Greek island. Treatment is a binary indicator of refugee arrival on the resident's island. Instrument is the logged distance between the resident's island and Turkey. Outcome is survey responses.

Figure 4 shows the causal effect of refugee arrival on each outcome variable as confidence intervals.
For missing values, authors claim to mark missing values for covariates with a binary indicator, and use Multiple Imputation by Chained Equations (MICE) to impute missing values for outcome variables. MICE assumes that data are missing at random, so their values can get predicted based on observed values. Then the authors create four broader components (asylum seeker attitudes, immigrant attitudes, Muslim attitudes, behavioral outcomes) to reduce false positive rate that comes from too many outcomes. They claim to use Principal Component Analysis to find the first principal component for the entire dataset.

However, when we replicate the results as Figure 4, we found two problems in the data provided in the figure replication code.
Firstly, 2,776 out of 221,616 data points are missing outcome values even after the imputation procedure was passed.
Secondly, 15,006 out of 61,560 principal component scores are missed even after the PCA procedure was passed. Upon further checking, we conclude that a lot of points do have outcome values that contribute to the missing components, hence there is no reason to leave them out of the PCA calculation.
These two missingness are problematic because (1) leaving out 24.3% of the dataset increases variance and reduces generalizability, (2) we cannot assume values are missing at random and the more we leave out, the more risk of biasing our results. Hence, the reliability of the causal effect direction and magnitudes come into question.

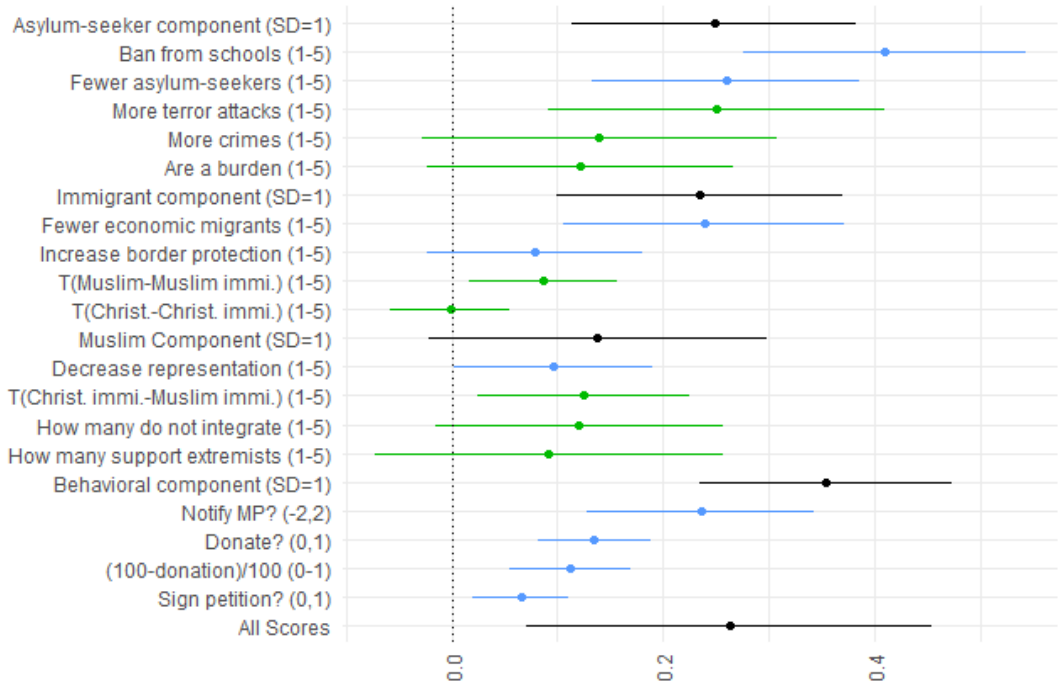


Original Figure 4.

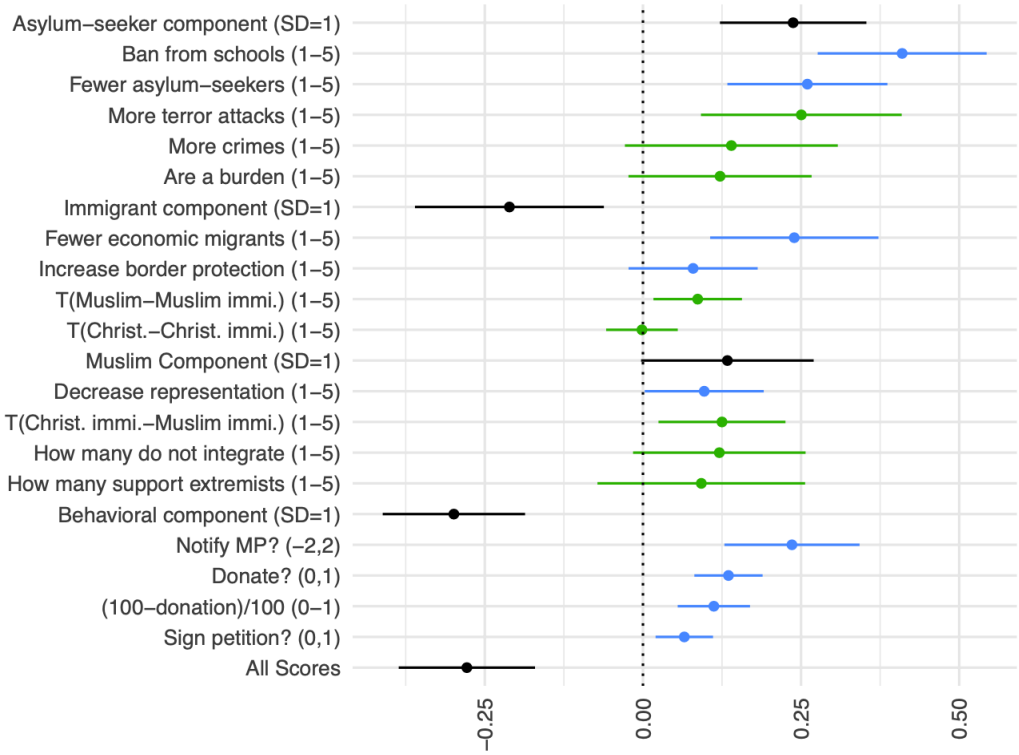
We extend the study by addressing the missingness problems. Our motivation is to correct the missing values and dimensional reduction and see how the results change. Building on the same approach of instrumental variable binary treatment multiple regression, first we modify missing data methods: we impute missing outcomes. Then, we compare imputation with dropping. Second, we modify dimensional reduction methods: we include all data in PCA. Then, we compare PCA with tSNE and UMAP.

EXTENSION:

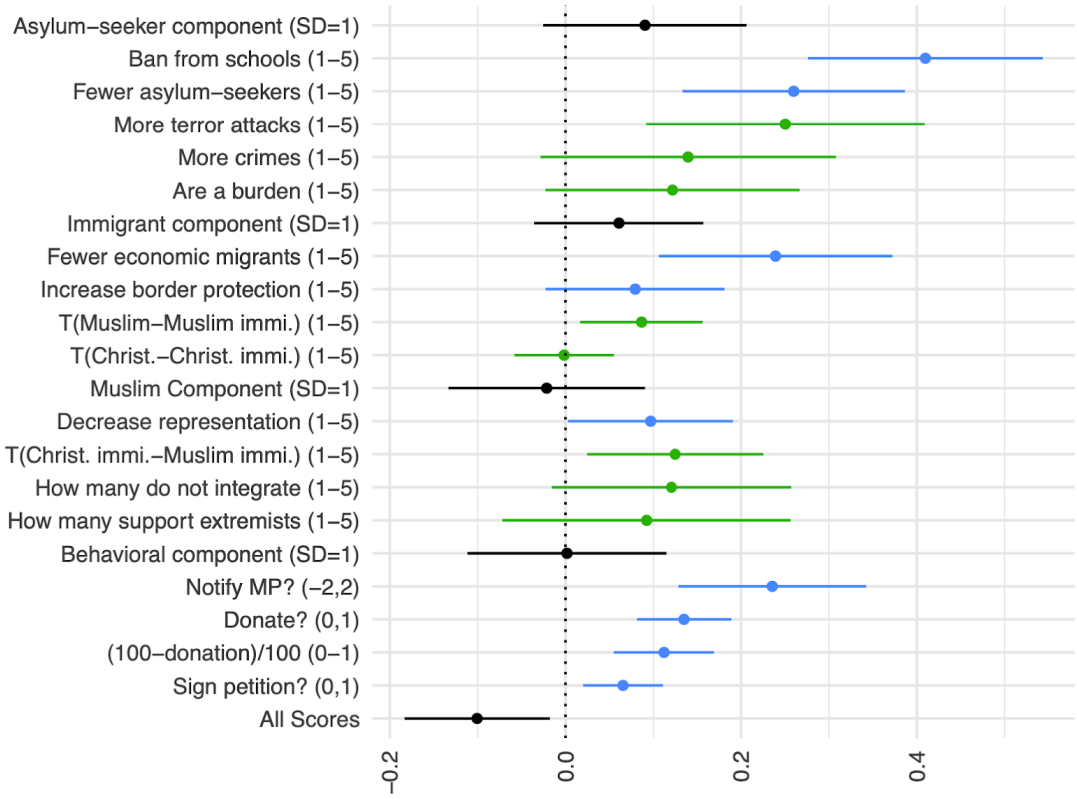
Fix missing values by multiple imputation by chained equations (MICE). MICE is a divide-and-conquer method for predicting missing values that assumes missingness is random.
To evaluate how much the dimensional reduction methods unbiased the data, we plot distributions of treatment vs control. We recompute estimates in Figure 4, using the new PCA, tSNE, UMAP.
PCA is a linear dimensionality reduction method. It is interpretable: we know what percentage of variance gets explained by a component. tSNE is a graph-based nonlinear method, which finds distance to neighbors instead of to the entire dataset. It is a stochastic greedy algorithm so everytime it gives a different convergence. It uses a cost function to minimize KL divergence, but it can't be used to evaluate how good the separation is.
All of the methods we try are unsupervised learning methods. PCA is linear while the other two are nonlinear, so by the bias-variance tradeoff PCA is likely less flexible but more generalizable, but the nonlinear ones save the intrinsic multidimensional nature. TSNE and UMAP differ by the latter preserving global structure too.
Methods need to fit the nature of the dataset. Our dataset has a large number of self reported categorical outcomes to reduce. They are likely interacting terms that are more like clusters than pairs.
We test the correlation between these dimensional reduction methods. The Pearson R between PCA and tSNE is 0.34 (only calculated for the first component), suggesting that the components generated by each method is positively correlated. However, correlation between UMAP and TSNE or UMAP and PCA are not presented as Pearson R is about 0, Lack of agreement implies that we cannot assume a functional reasoning behind the boundaries they create.
We can see that the direction of causal effect for behavioral outcome reversed after including all outcomes in dimensional reduction. Comparing between methods PCA tSNE UMAP, tSNE gives lowest magnitude estimates.



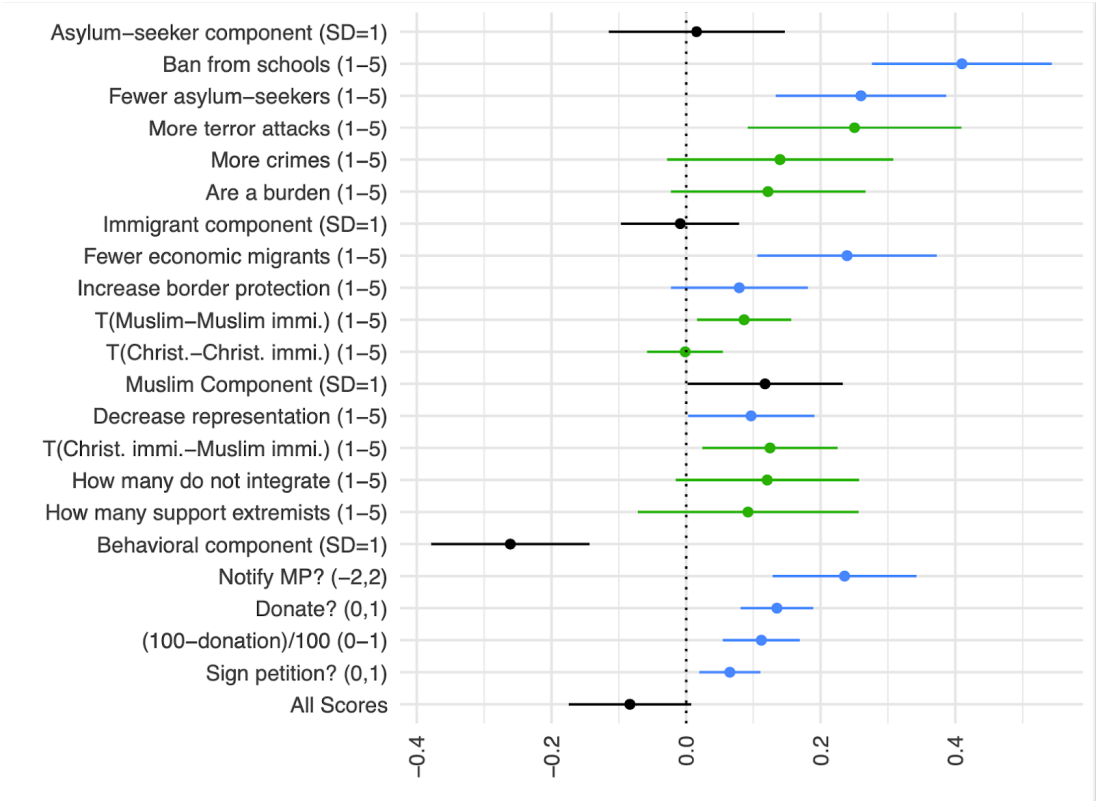
Replicated Figure 4 Using Original PCA



Replicated Figure 4 Using Our PCA



Replicated Figure 4 Using Our TSNE



Replicated Figure 4 Using Our UMAP

As we can see from the figures replicated using our estimates, most of the scores started to have a negative correlation with the treatment. It means that the imputations and correct dimensionality reduction (without having the unexpectedly missing values in the score metrics) led to the absolutely different confidence intervals. It is difficult to understand why it is so, and what it means in this particular case but we could extend the study and possibly correct it.

CONCLUSION:

We have shown that Hangartner et.al. (2019)'s causal effect estimates for component scores of resident hostility due to refugee arrival suffer from mishandled data missingness procedures. We produce estimates for several correction methods to contextualize the original study's results and observe a general reduction in effect of arrival than the original claim, with one component (behavior) consistently reversing direction of effect. That means, compared to what Hangartner said, having refugees arrive on an island doesn't cause as dramatic of an increased hostility toward refugees, immigrants, Muslims, and it may even decrease hostile behaviors.

In light of the latent hospitality which the authors try to address using the instrument of distance to Turkey, our corrections about non correlated dimensional reduction boundaries might imply that the instrument is not strong enough to explain enough covariance, hence we might need to further address covariate imbalance.

The limitations of our methods are that we have no universal metric for dimensional reduction accuracy since there are no labels.

In the future we can improve by collecting more data that has less missing values or has more empirical evidence to support assumptions behind each missing data fix and score-summary fix. For example, covariate balance of respondents and non-respondents can support the assumption about random missingness.

REFERENCES:
Hangartner, D., Dinas, E., Marbach, M., Matakos, K., & Xefteris, D. (2019). Does Exposure to the Refugee Crisis Make Natives More Hostile? *American Political Science Review, 113*(2), 442-455. doi:10.1017/S0003055418000813

We will replicate and extend Hangartner et.al. (2019). (link) (data) (methods plan)

We refer to a replication by Mohamed et.al. (2019). (link)

This is a decision memo to UNESCO because they deal with refugee crises. Or it is addressed to Hangartner et.al. because we improved their study and want them to check if ours is indeed improvement on theirs. We update our knowledge about the effect of exposure to refugee arrival on local hostility from Hangartner et.al. (2018) by debiasing their methods. We are both building on the survey data they collected and on the method of causal inference through multiple regression, instrumental variable, of data imputation, of dimensional reduction. Namely we contextualize their method for dealing with missing data and for dimensionality reduction and how these choices change the results. What we did is really important because both are subjective choices that might have biased the results which informs our understanding about how refugee crisis is changing the society and its actions, how that works under the hood.

First, we replicate figure 2 in the study, which shows the estimated effect of the instrument on the treatment. It is part of the Two stage least square regression (2SLS). We checked and annotated the replication code provided by the authors. Here is our replicated figure.

Then, we will extend the instrumental variable analysis. We use progressive matching to improve balance on the covariate of interest (distance to Turkey).

For theory supporting extension, we refer to Hangartner et.al. (2020) for a method of profiling compliers and non-compliers to measure the estimate validity. (link)

Hangartner et.al. (2019)'s research question is: What is the impact of refugee immigration on attitudes, policy preferences, political behaviors of European citizens?

For statistical methods, Hangartner uses an instrumental variable on an observational study to reduce exogeneity.

What is an instrumental variable analysis? **Instrumental variable** method addresses confounding by looking separately at compliers and noncompliers. It assumes exclusion restrictions in the dataset, which means that instrument is correlated with treatment but independent from outcome. We can use the Wald Estimator to estimate causal effect given an instrument.

Essentially, instrumental variable estimation is essential for leveraging partially exogenous treatment variables in observational studies. Without IV, when the treament

Step 2: replicate the IV estimation. discuss it with #critique, #biasidentification.

How the two stage regression works is just the ivreg function in the AER package.

```
reg.data <- survey_data %>% filter(mi_m == i)

iv.out <- ivreg(data = reg.data, reg.data[[j]] ~ treatment | logdist )

cluster.se <- sqrt(diag(vcovCL(iv.out, cluster = ~munid, type = "HC0",

                        target = NULL, inverse_var = FALSE)))

b.out <- rbind(b.out, iv.out$coef)

se.out <- rbind(se.out, cluster.se)
```

Figure 2 shows stage 1 estimation, which is the effect of assignment (distance to Turkey) on treatment (arrival).

Step 3: extend the IV. use #probability, #distributions, #regression, #confidenceinterval, #gapanalysis, #networks, #levelsofanalysis, #mutiplecauses, #emergentproperties.

The limitations of this study is the validity of the instrument. We might argue that distance to Turkey can increase hostility because they are the borders; that the assignment is not balanced in treatment and control; that most control units had no possibility of receiving treatment.

To improve Hangartner et.al. (2019) we can check instrument validity. We want to check if there is an imbalance of an unobserved characteristic between the treatment and control groups which may bias the results.

How close islands receiving and not receiving treatment must be in order to be seen as comparable? how much balance on covariates must exist between treatment and control groups for critical assumptions of causal inference to hold.

A more restrictive specification would increase the reliability of our study design, but oftentimes also decrease our confidence about a result (because the confidence interval widens and the effect size shrinks).

PCA

Original

Item nonresponse and missing data

Data for the treatment indicators is available for all observations. For missing covariate values, we will include a separate binary indicator. For missing outcome values, we will employ multiple imputation using chained equations (MICE) before conducting the PCA. We generate five imputed datasets, estimate the PCA and models on each of them and combine the results. For the imputation we condition on the entire survey dataset.

Lastly, for each of the four sets of outcomes, we built a summary scale that combines the different measures by extracting the first component of a polychoric principal component analysis (PCA) (Olsson 1979). Appendix Section A.3 provides details on the PCA and shows that for each scale, the first component explains a large share of the total variance. Averaging responses across multiple items measuring the same latent construct not only reduces bias and variance from random measurement error that is common in survey research (Hainmueller, Hangartner, and Pietrantuono 2017), but also alleviates worries about false-positive significance tests by focusing the analysis on a few main outcomes.

The sampling frame, questionnaire, coding rules for covariates and outcomes (including the summary scales extracted from the PCA), and statistical analyses were all prespecified in a pre-analysis plan posted to the Political Science Registered Studies Dataverse before we conducted the survey and can be retrieved at doi: 10.791/V/FEUEH.

Student

Note that our PCA components (which can be seen in the black lines in Figures 2 and 4b-7b) were produced using Stata, using the original authors' code. We then imported the PCA components into R to produce the regression analysis and corresponding figures.

Original Paper

1. **liquidtrees**13 Dec 2020

   in Public

   Our instrument is the logarithm of the shortest path (henceforth distance) between Turkish coast and the population center of a municipality (typically an island), which we calculate using Google Maps.

   how did they use the instrument on the statistical analysis? did they match same instrument different treatment?

2. **liquidtrees**13 Dec 2020

   in Public

   While some of the estimates are slightly larger for respondents with more hostile views, these differences are not always statistically significant.

   the quantile IV regression showed that the covariate (preexisting hostility) did not matter.

3. **liquidtrees**13 Dec 2020

in Public

Despite the difference in the empirical strategy and the focus on highly local variation in exposure to the refugee crisis, where spillover effects on island residents slightly farther away are likely, the findings are remarkably similar to our main estimates identified from between-island variation.

the authors find, by using different distances as Z, that hostility always increases, even at very small difference in Z.

4. **liquidtrees**13 Dec 2020

in Public

If there is an unobserved confounder correlated with distance, the effect estimates for the subsample of islands within a 255-kilometer (the midpoint of our distance variable) perimeter of Turkey should be different from the estimates from the full sample.

how similar is similar? we should check E4 to see if the result is good enough to ignore this covariate.

5. **liquidtrees**13 Dec 2020

in Public

To estimate the effect of exposure to the refugee crisis, we fit two-stage least squares (2SLS) models in which we regress the outcome on our binary exposure indicator controlling for respondent demographic characteristics.

they regressed outcome on exposure indicator, which includes effect by instrument. alternatively they could've used other regression models.

6. **liquidtrees**13 Dec 2020

in Public

Averaging responses across multiple items measuring the same latent construct not only reduces bias and variance from random measurement error that is common in survey research (Hainmueller, Hangartner, and Pietrantuono Reference Hainmueller, Hangartner and Pietrantuono2017), but also alleviates worries about false-positive significance tests by focusing the analysis on a few main outcomes.

authors argue that their survey data is improved by PCA. to improve data we can remove outliers, match, synthesize.

7. **liquidtrees**13 Dec 2020

in Public

The robustness section below shows that all results also hold with alternative weighting strategies

how do weights on responses change the conclusion? how can we tweak this?

8. **liquidtrees**13 Dec 2020

in Public

in order to remove any imbalances and increase the efficiency of our estimates, we use multiple imputation to address missing answers

because, people who don't respond might also be less hostile.

9. **liquidtrees**13 Dec 2020

in Public

Table 1 provides the estimates from two-stage least squares regressions, our main specification further discussed below, of the placebo outcomes on a binary indicator for refugee arrivals, instrumented with distance to the Turkish coast.

authors prove instrument validity by showing that placebo tests do not show that the instrument changes other covariates.

10. **liquidtrees**13 Dec 2020

in Public

This instrumental variable approach allows us to resolve the selection bias associated with refugees choosing their arrival island based on preexisting levels of hostility toward outgroups.

the authors argue that proximity overpowers hospitality in a refugee's choice

11. **liquidtrees**13 Dec 2020

in Public

social threat can explain how the chaotic management of refugee flows activates exclusionary reactions to immigrants and ethnic minorities, even in a context where these flows are temporary and transient.

this hypothesis makes sense. the mechanism behind the causal effect would then be: see management failures like detained people, visible outgroup

12. **liquidtrees**13 Dec 2020

in Public

All these factors are believed to catalyze anti-immigrant sentiment

does the instrumental variable account for the difference in how much the financial crisis and management affect hostility?

13. **liquidtrees**13 Dec 2020

in Public

geographical proximity and the fact that they often belong to the same administrative unit ensures that islands with and without arrivals are identical across many observable and unobservable characteristics

so, the authors use distance to turkey like this: there are islands that have the same distance, but different amounts of refugee.

14. **liquidtrees**13 Dec 2020

in Public

This solves the aforementioned selection issue by focusing the analysis on the exogenous part of the variance in refugee arrivals that is solely driven by proximity to Turkey and ignoring the endogenous part that is driven by preexisting differences in attitudes toward refugees.

is an instrument the same as an encouragement assignment? but, wouldn't areas closer to Turkey also tend to be more hostile to refugees, because they are on the border?

15. **liquidtrees**13 Dec 2020

in Public

This latent hospitality toward refugees is hard to measure empirically but will typically confound the relationship between refugee arrivals and outcome measures of hostility

How do the authors address the confounder, hospitality?

Student Paper

1. **liquidtrees**13 Dec 2020

   in Public

   Instrumental variables assumptions are notoriously difficult to satisfy,and it may be that none of these methods suffice to retain confidence in the original findings. On the other hand, it may be that some restrictions presented here go too far.

   how might we determine an authoritative specification?

2. **liquidtrees**13 Dec 2020

   in Public

   Noting the balance-sample size trade off that exists in matching analysis (King, Lucas andNielsen, 2017), we do not provide a single revised matched estimate. Instead, we use the algorithm proposed by King, Lucas and Nielsen (2017) to rerun the instrumental variables analysis using eight different post-matching subsets of the original data, successively pruning 150 observations at a time in an attempt to improve balance

   they match several times to prune the dataset

3. **liquidtrees**13 Dec 2020

   in Public

   assumptions of two-stage least squares regression

what are the assumptions? (unconfoundedness?) is this a typical IV procedure?

4. **liquidtrees**13 Dec 2020

in Public

authors' original analyses while selecting only data which we view as lying plausibly within the potential treatment group

although IV is less robust in this subset, if we assume it holds, then the causal effect is less. is it still significant?